

SEMESTER END EXAMINATIONS - JANUARY 2026

Program	: B.E. - Information Science and Engineering	Semester	: V
Course Name	: Computer Networks	Max. Marks	: 100
Course Code	: 23IS53	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Define Data Communication. Explain fundamental characteristics of data communication systems. CO1 (06)
 - Define Nyquist bit Rate and Shannon capacity. Calculate the propagation time and transmission time for a 2.5 KB message if the bandwidth of the network is 10Gbps? Assume that the distance between the sender and the receiver is 15,000km and the light travels at 2.4×10^8 m/s. CO1 (06)
 - Discuss the layered architecture of OSI model with a neat diagram. CO1 (08)
- Represent the given sequence 01001110 using NRZ-L, Manchester, AMI and Pseudoternary line coding schemes, convert the sequence 010011 to digital signal. CO1 (08)
 - Describe briefly the various causes of transmission impairment with necessary diagrams. CO1 (09)
 - The loss in a cable defined in decibels/km(dB/km). If the signal at beginning of a cable with -3 dB/km has a power of 3 mW. What is the power of the signal at 10km? CO1 (03)

UNIT - II

- Define cyclic codes and linear block codes. Given the dataword 101011 and the divisor 10011, generate the codeword using CRC polynomial division. CO2 (07)
 - Illustrate the different fields of ARP packet with a neat diagram. CO2 (07)
 - Compare bit stuffing and byte stuffing with suitable examples. CO2 (06)
- What is transmission impairment? Briefly explain its causes with necessary diagrams. CO2 (07)
 - Calculate the Internet check sum for the text "ISE DEPARTMENT" at sender and receiver side (assume the following hexadecimal ASCII values: A=41, D=44, E=45, I=49, M=4D, N=4E, P=50, R=52, S=53, T=54, space=32). CO2 (06)
 - In a pure ALOHA network, 300 nodes, each generates 3 frames/min. Each frame transmission time = 40ms. Find the offered load G on the channel. Also differentiate Pure ALOHA from Slotted ALOHA. CO2 (07)

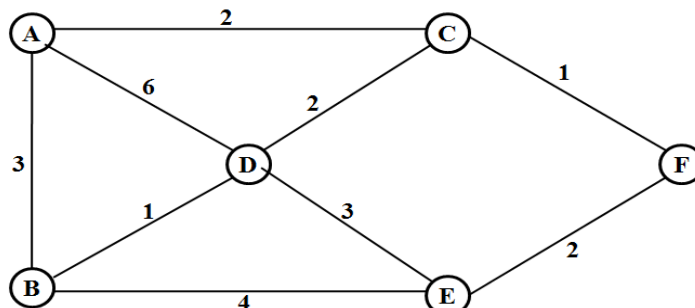
UNIT - III

- An IPV4 datagram has arrived with the following information in the header (in hexadecimal) CO3 (08)

45 00 00 54 00 03 58 50 20 06 00 00 7C 4E 03 02 B4 0E 0F 02

- Calculate the header checksum.
- Are there any options?
- What is the size of the data?
- How many more routers can the packet travel to?

- b) Describe different decision-making strategies used by distance vector multicast routing (DVMRP) protocol with necessary diagram. CO3 (08)
- c) Illustrate the different open loop congestion control techniques. CO3 (04)
6. a) Applying Link State Routing and calculate the shortest path from the root node A to all other nodes in the following network configuration. CO3 (08)



- b) What are the three major restrictions imposed on classless address blocks? An Organization is granted the block 211.17.180.0/24. The administrator wants to create 32 subnets.
 (i) Find the subnet mask
 (ii) Find the number of addresses in each subnet
 (iii) Find the first and last address of subnet 3
 (iv) Find the network address of subnet 25
- c) Briefly discuss the different error reporting messages in ICMP protocol. CO3 (05)

UNIT- IV

7. a) With a neat header diagram, explain the different fields of an IPV6 datagram. CO4 (08)
- b) Giving the flow diagram, Illustrate the behavior of Stop-and-Wait Protocol at transport layer. CO4 (08)
- c) Differentiate between IPV4 and IPV6 headers. CO4 (04)
8. a) Justify the statement "Size of the send window must be less than 2^m in Go-Back-N ARQ protocol" with necessary diagrams. CO4 (08)
- b) Briefly describe the different strategies to achieve transition from IPV4 to IPV6. CO4 (06)
- c) Differentiate between connection oriented service and connectionless service. CO4 (06)

UNIT - V

9. a) Explain the handling of congestion in reno TCP using necessary diagrams. CO4 (10)
- b) Mention some typical applications of UDP. CO4 (04)
- c) Discuss the use of handshaking protocol in TCP for establishing the connection. CO5 (06)
10. a) With a neat diagram, explain the different fields in a TCP segment. CO5 (10)
- b) Discuss the different mechanisms used by DNS in name-address resolution. CO5 (10)
